

Confidex and Sasken
proudly present the Dynamic Parts Authentication™

CURE FOR THE COUNTERFEITING PANDEMIC IN THE AUTOMOTIVE INDUSTRY

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JUNE 2017



INTRODUCTION

Apparel, accessories, music, software, medications, cigarettes, consumer goods, toys, electronics, automotive and aircraft parts...

Counterfeits are everywhere in industry, and their manufacturers are exploiting trusted online platforms—such as Amazon, eBay and Facebook—to distribute their products. The counterfeiting business is worth £1.18 trillion, or 10% of global trade**, and is growing at an annual rate of around 20%.

In other words, the counterfeiting business is a pandemic.

Counterfeiting attracts organized crime, as the risk of prosecution is low and the profit potential is enormous. Counterfeiting is also reportedly linked to piracy and terrorism, with fake and unauthorized goods being sold to raise funds and launder money.



One in six products sold online can be proven to be counterfeit.*

*SOURCE: <https://www.netnames.com/online-brand-protection/>

**SOURCE: <https://www.netnames.com/assets/shared/whitepaper/pdf/NetNames-Counterfeiting-Report-A4-2015.pdf>

Today, it is difficult to tell a counterfeit from the genuine product. As counterfeits are designed to be as perfect replicas as possible, not even car mechanics and other professionals can recognize a counterfeit when they see one.



The annual cost of counterfeiting to the automotive industry is £28 billion.*

*SOURCE: <https://www.netnames.com/assets/shared/whitepaper/pdf/NetNames-Counterfeiting-Report-A4-2015.pdf>

COUNTERFEITING IN AUTOMOTIVE INDUSTRY

Spare parts are crucial to keep cars running safely and reliably and ensure high performance. Counterfeit spare parts are deceptive and dangerous; for example, brake pads that look like genuine, high-quality products may contain sawdust instead of friction material, and the filter element in an oil filter may be made of old newspapers.

The trade in counterfeit components involves an extensive selection of products, from automotive glass, oil and air filters, wheels, brakes, and suspension and coupling mechanisms to seat belts.

Driver and passenger safety simply cannot be compromised. Profit margins must never be based on fake parts.

Price may be a key point for a service facility seeking to increase profits, but it is important to remember that the ramifications of driving a car with unreliable brakes are very different from those of wearing counterfeit sneakers, for example.

The automotive industry is estimated to lose £28 billion annually** because of counterfeiting. Aftermarket parts manufacturers find this estimate far too conservative. According to Business Wire, the counterfeit car parts business was worth £26 billion (\$45 billion) in 2013.



Counterfeit parts
cause 50% of
motor vehicle deaths
in Saudi Arabia.*

*SOURCE: <https://www.netnames.com/assets/shared/whitepaper/pdf/NetNames-Counterfeiting-Report-A4-2015.pdf>

**SOURCE: <http://automotivelogistics.media/intelligence/16979>

WHY SHOULD I CARE?

Counterfeiters destroy brand value. According to the International AntiCounterfeiting Coalition (IACC), we should also take counterfeiting seriously for many other reasons:**

① IT'S DANGEROUS

Counterfeit goods are often made using cheap, substandard, dangerous components that risk consumer health and safety.

② IT COULD COST YOU

Purchasing from counterfeit websites puts you at risk for identity theft and credit card fraud. Downloading or streaming from illegal websites also puts you at risk for malware that can steal your personal or credit card information.

③ IT'S ILLEGAL

Counterfeiting is illegal, and purchasing counterfeit products supports illegal activity.

④ IT COSTS YOUR CITY

Counterfeiters do not pay taxes, which means less money for the schools, hospitals, parks, social programs and other vital needs in your city.

⑤ IT SUPPORTS CHILD LABOR

Counterfeiters do not pay their employees fair wages or benefits. They often use forced or child labor and provide poor working conditions.

⑥ IT SUPPORTS ORGANIZED CRIME

The profits from counterfeiting have been linked to funding organized crime, trafficking drugs and supporting terrorist activity.

⑦ IT HURTS LEGITIMATE COMPANIES

Legitimate manufacturers invest substantial resources in research and product development and in building a reputation for quality among consumers. Counterfeiters seek to profit from this work and reputation. The sales and profits lost because of this unfair competition translate directly into lower wages, lost jobs, and higher prices for consumers.



58% of all counterfeit car parts are bought online.*

*SOURCE: <https://www.pwc.co.uk/assets/pdf/anti-counterfeiting-consumer-survey-october-2013.pdf>

**SOURCE: <http://www.iacc.org/resources/about/what-is-counterfeiting>

GENUINE PARTS ARE VERIFIED IN THE CLOUD

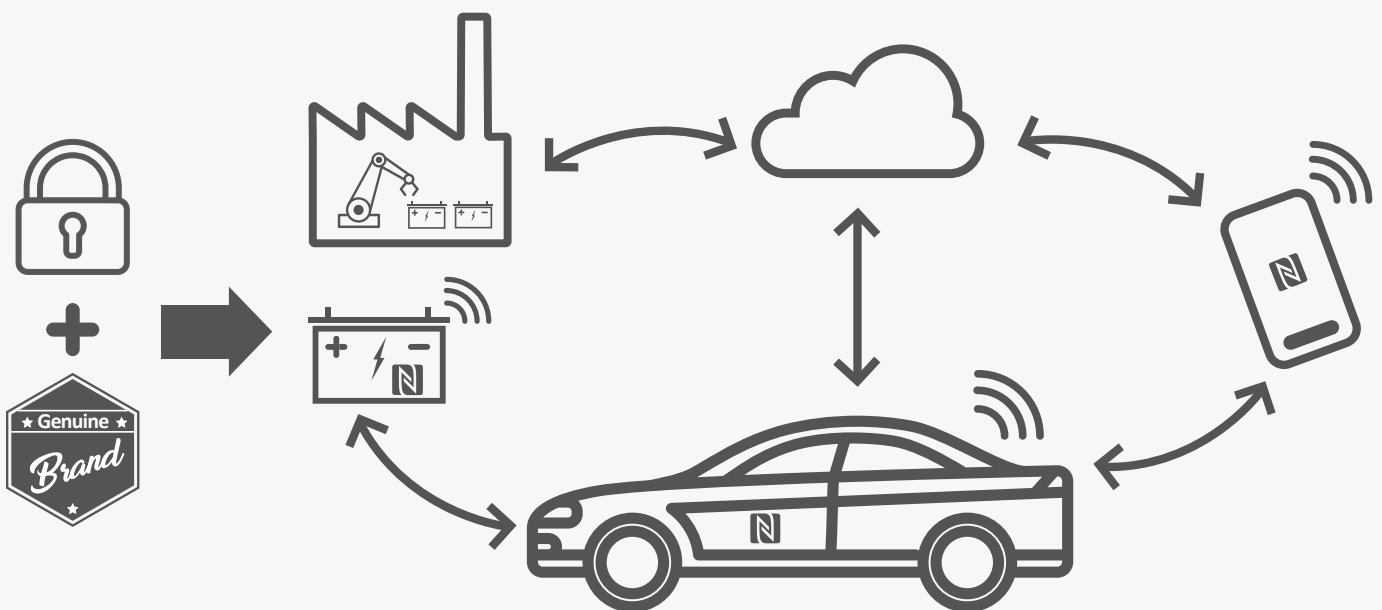
For the automotive industry, the good news is they now have a solution for protecting companies and their shareholders and customers. Confidex and Sasken proudly present the **Dynamic Parts Authentication™** (DPA™) solution.

The DPA solution helps automotive manufacturers ensure that various stakeholders and even the car itself can verify whether critical spare parts are genuine. This cloud-based solution authenticates spare parts wirelessly using secure near field communication (NFC) technology.

In a car, the DPA solution uses integrated NFC readers, the car's internal network, and cloud-connected telematic control units (TCU)

to check ID-related information about NFC-tagged parts from a cloud-based authentication database. Car parts are linked to the DPA system using **Confidex Links NFC™** tags that can be customized for different types of spare parts. A secure authentication ID is embedded in the part's NFC tag at the time of manufacturing, and the related secure keys are saved to the automotive OEM's or Tier 1 manufacturer's DPA cloud.

Consumers can check spare parts using NFC-enabled smartphones or get automated app notifications if a part is being illegally removed from the car. The DPA solution can make use of online information from cloud databases, such as those of the OEM or the Tier 1 parts manufacturer, or via trusted third parties.



If there is reason to believe a part is counterfeit, the DPA solution can send automated notifications via multiple channels to the owner of the car or the maintenance and warranty database of the parts manufacturer or the OEM brand, among other recipients.

The DPA solution can also be used to dynamically tune and optimize car settings directly from genuine, value-added spare parts. For example, an authentic high-performance battery can optimize the performance of an electric vehicle in winter conditions, and a genuine racing air filter can optimize the settings related to the air intake in a car engine.



Near field communication (NFC) is a standards-based short-range wireless connectivity technology. It simplifies transactions and makes it easier to exchange digital content and connect electronic devices with a touch.

WHY NFC?

- Makes use of the existing infrastructure of close 3 billion NFC smartphones
- Ensures accuracy in close-distance reading
- Offers advanced security levels

BUSINESS CASES WITH DPA:

①

Protecting consumer safety against counterfeits and gray-market copies without extra effort from the consumer

②

Clarifying liabilities related to warranties and maintenance in the spare parts business of the automotive industry

③

Protecting and enhancing the car manufacturer's brand reputation by verifying the genuine parts supply

④

Creating value-added business for manufacturers of genuine spare parts through dynamic settings functionality

⑤

Improving the accuracy of the maintenance and repair history of individual cars

⑥

Increasing visibility across the automotive manufacturing and supply chains through automatic identification of parts

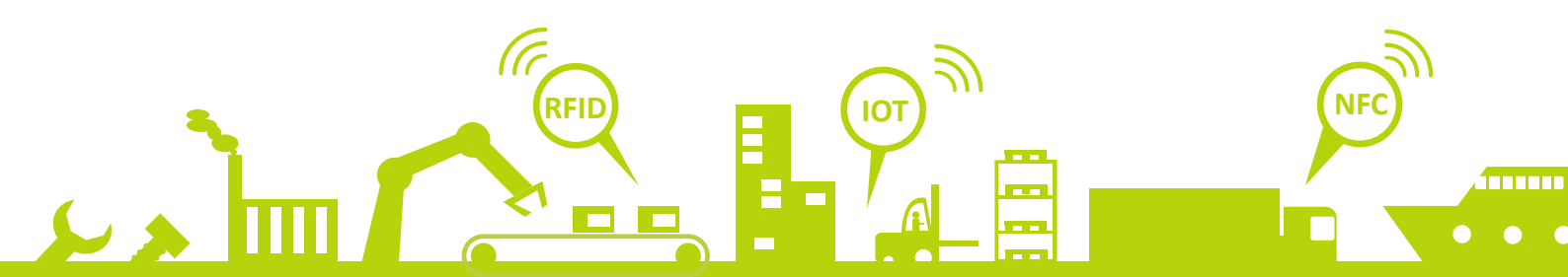
DID YOU KNOW THAT OVER 2 MILLION PEOPLE DRIVE VOLVOS THAT HAVE BEEN BUILT AROUND OUR RFID TAGS?

Confidex has been creating the Internet of Things (IoT) for over 13 years. We link cars and put the radio frequency identification (RFID) tags through the painting process at Volvo Cars. We are trusted by the world's largest car manufacturers, and more than 30 other major companies (over €1 billion) link with us.

Wireless identification is being used widely across the automotive supply chain, which is known for its extremely high quality standards. Confidex RFID technology is used to automatically identify assets throughout the manufacturing and logistics processes.



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WATCH VIDEO and find more about **Dynamic Parts Authentication™** and how it can secure your brand, visit www.confidex.com/mydpa

In cooperation with

